

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Rising Waters - A Flood Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S. NO	Limitations	Impact	Priority to Address (High / Medium / Low)
1	Local deployment only	Limited user access	High
2	Historical dataset	Prediction depends on dataset quality	High
3	No live weather integration	Cannot provide real-time prediction	Medium

Scalability Plan:

S. NO	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on IBM Cloud with multiple users
2	Data Storage	Static CSV dataset	Store data in cloud database
3	Performance	Single Flask server	Load balancing and optimized deployment
4	Security	Basic local application	Add authentication and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time weather API integration	3 Months	More accurate predictions
Phase 2	Cloud deployment (IBM Cloud)	6 Months	Improved accessibility
Phase 3	Mobile application	9 Months	Easier user access
Phase 4	SMS & Email flood alerts	12 Months	Faster disaster response